Hong Ong

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Professional Summary

Software Engineering graduate from UTS with First Class Honours and nearly a year of production experience. Sole software engineer on OnlyFacts' Ruby on Rails/PostgreSQL climate data platform, and developer on The a2 Milk Company's Astro/TypeScript websites across 8 markets. Full-stack across data pipelines, REST APIs, CMS-driven components, localisation and automated testing with Vitest and RSpec.

Work Experience

Software Developer (Contract) June 2026 – Present
The a2 Milk Company *Sydney, NSW*

- Developed features for multi-market Astro 5 SSR sites (AU, NZ, SG, HK, VN, CA, US, China) with TypeScript, Kontent.ai (headless CMS), Alpine.js, Algolia and Tailwind CSS on AWS
- Shipped 125+ commits across 3 production repositories, delivering 12+ reusable CMS-driven components with documentation and Vitest unit and smoke tests
- Built a modular article system of 7 CMS-driven components for a health-professional research hub, launching 6 articles to production
- Added Vietnamese and Chinese language support to UI labels, filters and layouts on regional sites, and integrated Osano cookie consent and GTM

Full-Stack Engineer (Contract) January 2026 – Present
OnlyFacts *Sydney, NSW*

- Managed all engineering on the Ruby on Rails/PostgreSQL climate data platform as sole software engineer since mid-2026, working with a 4–6 person data team
- Replaced Highcharts maps with a custom D3.js engine (council-area and world maps, hexbin view, zoom, legend) and refactored 1,300+ lines of legacy code
- Built web scrapers and daily and quarterly import jobs for 7 data sources, with email alerts on new releases that replaced the data team's manual checks
- Shipped a Cloudflare Worker contact API (validation, Google Sheets logging, email alerts, CORS allowlist) and GA4 event tracking for the live React/Vite OnlyFacts Studio site

Software Engineering Intern *November 2025 – January 2026*

- Prototyped interactive D3.js/TopoJSON maps with tooltips and data normalisation, later extended into the production map engine

Software Engineering Consultant May 2024 – February 2025
Optik Consultancy (Philips Dynalite) *Sydney, NSW*

- Led a 4–5 person team of software and mechanical engineers to deliver client automation solutions on time and within budget
- Programmed OMRON TM5 cobot using TM FLOW to automate relay/terminal placement, cutting production time by 40% and error rates by 60%
- Reviewed 5 software projects for progress against project goals and quality standards

Education

University of Technology Sydney Sydney, NSW
Bachelor of Engineering (Honours), Software Engineering *February 2022 – December 2025*

- First Class Honours | WAM: 87.10 (High Distinction Average) | GPA: 6.59/7.0 | Dean's List (3 consecutive years) | Undergraduate Academic Excellence International Scholarship
- Honours Capstone (HD, showcase nominee): built a C++/Python AST-based plagiarism detector that flagged 8 confirmed misconduct cases over 406 submissions, 73% faster with 73% less storage than text matching
- Relevant Coursework: Database Fundamentals (100), Cloud Computing & SaaS (97), Software Innovation Studio (96), Software Analysis Studio (94), Software Architecture (93), Systems Testing and Quality Management (89)

Projects

VR Career Exploration: Lifeguard Simulation | *Unity, C#, XR Toolkit* August 2025 – December 2025

- Built a VR training simulation in Unity/C# with XR Interaction Toolkit and a controller-based selection system with haptic, audio and visual cues that reached 95% selection accuracy for first-time users at a live showcase
- Developed an NPC system with randomised spawning and optimised it for VR headsets with object pooling and fewer draw calls, cutting frame drops by about 50% (Unity profiler)
- Led a 7-person Agile team to People's Choice and Best Trailer (class showcase), running code reviews and technical QA across 3D modelling, animation and UI

Thrive X Mobile App Prototype | *Flutter, Dart, Firebase, Google Gemini API* February 2025 – April 2025

- Built a cross-platform Flutter (Dart) mobile app with the Google Gemini API for AI-driven skill assessments and personalised feedback
- Implemented real-time data sync via Firebase (30% faster load times), RESTful API integration, and user authentication/profile management

Leadership Activities

Treasurer November 2022 – June 2024

UTS Malaysian and Singaporean Students' Association (MASSA) *Sydney, NSW*

- Managed a $40,000+ budget across 20+ events and consistently broke even or made a profit

Skills

Languages: TypeScript, JavaScript, Ruby, SQL, Python, C++, C#
Frontend: Astro, React, Alpine.js, Stimulus/Turbo, D3.js, Tailwind CSS, Flutter
Backend: Ruby on Rails, REST APIs, Cloudflare Workers, background jobs (Solid Queue, Heroku), web scraping
Data & CMS: PostgreSQL, Firebase, Kontent.ai (headless CMS), Algolia
Cloud: AWS (CloudFront, Elastic Beanstalk, CodeBuild, S3), Heroku, Cloudflare, GCP
Testing: Vitest, RSpec (unit & integration)
DevOps & Tools: Git, CI/CD (GitHub Actions, AWS CodeBuild), Docker Compose, Agile/Scrum, Claude Code
Analytics: GA4, GTM, Osano cookie consent
Spoken Languages: English, French (fluent), Vietnamese, Mandarin, Cantonese (basic)